- Coal is used extensively as a fuel in most parts of the world.
- Burning coal produces about 15 billion tonnes of carbon dioxide each year.
- Attempting to use coal without adding to atmospheric carbon dioxide levels is a major technological challenge.
- The greatest challenge is bringing the cost of this down sufficiently for 'clean coal' to compete with nuclear

Privacy - Terms

power on the basis of near-zero emissions for base-load power.

- There is typically at least a 20% energy penalty involved in 'clean coal' processes.
- World R&D on CCS exceeded $1 billion per year over 2009 to 2013, then fell sharply.
- The term 'clean coal' is increasingly being used for supercritical coal-fired plants without CCS, on the basis that CO_2 emissions are less than for older plants, but are still much greater than for nuclear or renewables.

Some 26% of primary energy needs are met by coal and 37% of electricity is generated from coal, compared with 23% for natural gas. Coal is the world's most abundant and widely distributed fossil fuel source. Globally about 2000 GWe of coal-fired generation capacity is operating and another 500 GWe is expected online by 2030.

However, each year burning coal produces over 15 billion tonnes of carbon dioxide (CO_2), which is released to the atmosphere, most of this being from power generation.

Development of new 'clean coal' technologies is attempting to address this problem so that the world's enormous resources of coal can be utilised for future generations without contributing to global warming. Much of the challenge is in commercialising the technology so that coal use would remain economically competitive despite the cost of achieving low, and eventually 'near-zero', emissions. The technologies are both costly and energy-intensive.

As many coal-fired power stations approach retirement, their replacement gives much scope for 'cleaner' electricity. Alongside nuclear power and harnessing renewable energy sources, one hope for this is via 'clean coal' technologies, such as carbon capture and sequestration, also called carbon capture and storage (both abbreviated as CCS) or carbon capture, use and storage (CCUS). It involves the geological storage of CO_2, typically 2-3 km deep, as a permanent solution. However in its *Energy Technology Perspectives 2014* the International Energy Agency (IEA) notes: “CCS is advancing slowly, due to high costs and lack of political and financial commitment.” In Energy Technology Perspectives 2020, CCUS is fairly low profile, and in the Sustainable Development

Scenario about 2 billion tonnes per year is captured from coal burning by 2070 (of 10 Gt/yr total). In 2020 about 40 million tonnes of CO_2 per year was being captured and sequestered from all sources, with about the same CCS capacity under construction. See also IEA webpage on CCUS.

Consequently the term 'clean coal' is increasingly being used for supercritical and ultra-supercritical coal-fired plants without CCS, running at 42-48% thermal efficiency. These are also known as high-efficiency low-emission (HELE) plants. The capital cost of ultra-supercritical (USC) HELE technology is 20-30% greater than a subcritical unit, but the higher efficiency reduces emissions and fuel costs to about 75% of subcritical plants. A supercritical steam generator operates at very high temperature (about 600 °C) and pressures (above 22 MPa), where liquid and gas phases of water are no longer distinct. In Japan and South Korea about 70% of coal-fired power comes from supercritical and ultra-supercritical plants.

IEA report in 2020

In 2020 the IEA published a report on CCUS in its *Energy Technology Perspectives 2020* series. It said that CCUS "will need to form a key pillar of efforts to put the world on the path to net-zero emissions… It is the only group of technologies that contributes both to reducing emissions in key sectors directly and to removing CO_2 to balance emissions that cannot be avoided" (*i.e.* including direct removal of CO_2 from the atmosphere). Summarizing the report, the IEA said: "CCUS can be retrofitted to existing power and industrial plants that could otherwise emit 600 billion tonnes of CO_2 over the next five decades." In the IEA's Sustainable Development Scenario, CCUS accounts for nearly 15% of the cumulative reduction in emissions by 2070 (compared with its Stated Policies Scenario). While CCUS facilities have been operating for decades in certain industries, "they are still a work in progress in the areas that need them most."

Managing waste from coal

Burning coal, such as for power generation, gives rise to a variety of wastes which must be controlled or at least accounted for. So-called 'clean coal' technologies are a variety of evolving responses to late 20th century environmental concerns, including that of global warming due to carbon dioxide releases to the atmosphere.

However, many of the elements have in fact been applied for many years, and they will be only briefly mentioned here:

- Coal cleaning by 'washing' has been standard practice in developed countries for some time. It reduces emissions of ash and sulfur dioxide when the coal is burned.
- Electrostatic precipitators and fabric filters can remove 99% of the fly ash from the flue gases – these technologies are in widespread use.
- Flue gas desulfurization reduces the output of sulfur dioxide to the atmosphere by up to 97%, the task depending on the level of sulfur in the coal and the extent of the reduction. It is widely used where needed in developed countries.
- Low-NOx burners allow coal-fired plants to reduce nitrogen oxide emissions by up to 40%. Coupled with re-burning techniques NOx can be reduced 70% and selective catalytic reduction can clean up 90% of NOx emissions.
- Increased efficiency of plant – up to 46% thermal efficiency now (and 50% expected in future) means that newer plants create less emissions per kWh than older ones. See Table 1.
- Advanced technologies such as integrated gasification combined cycle (IGCC) and pressurized fluidized bed combustion (PFBC) enable higher thermal efficiencies still – up to 50% in the future.
- Ultra-clean coal (UCC) from new processing technologies which reduce ash below 0.25% and sulfur to very low levels mean that pulverised coal might be used as fuel for very large marine engines, in place of heavy fuel oil. There are at least two UCC technologies under development. Wastes from UCC are likely to be a problem.
- Gasification, including underground coal gasification (UCG) in situ, uses steam and oxygen to turn the coal into carbon monoxide and hydrogen.
- Sequestration refers to disposal of liquid carbon dioxide, once captured, into deep geological strata.

Some of these impose operating costs and energy efficiency loss without concomitant benefit to the operator, though external costs

will almost certainly be increasingly factored in through carbon taxes or similar which will change the economics of burning coal.

However, waste products can be used productively. In 1999 the EU used half of its coal fly ash and bottom ash in building materials (where fly ash can replace cement), and it used 87% of the gypsum from flue gas desulfurisation.

Carbon dioxide from burning coal is the main focus of attention today, since it is implicated in global warming, and a series of international agreements require that emissions decline, notwithstanding increasing energy demand.

CCS technologies are in the forefront of measures to enjoy 'clean coal'. CCS involves two distinct aspects: capture, and storage.

The energy penalty of CCS is generally put at 20-30% of electrical output, though since no full commercial systems on power stations are yet in operation, this is yet to be confirmed. US and European figures below suggest a small or even negligible proportion.

Table 1. **Coal-fired power generation, thermal efficiency**

country	Technology	Efficiency	Projected efficiency with CCS
Australia	Black ultra-supercritical WC	43%	33%
	Black supercritical WC	41%	
	Black supercritical AC	39%	
	own ultra-supercritical WC	35%	27%
	own supercritical WC	33%	
	own supercritical AC	31%	
Belgium	Black supercritical	45%	
China	Black supercritical	46%	
Czech Republic	own PCC	43%	38%
	own ICGG	45%	43%
Germany	Black PCC	46%	38%
	own PCC	45%	37%
Japan, Korea	Black PCC	41%	
Russia	Black ultra-supercritical PCC	47%	37%
	Black supercritical PCC	42%	

South Africa	Black supercritical PCC	39%	
USA	Black PCC & IGCC	39%	39%
USA (EPRI)	Black supercritical PCC	41%	

OECD Projected Costs of Generating Electricity 2010, Tables 3.3. PCC= pulverised coal combustion, AC= air-cooled, WC= water-cooled.

Capture & separation of CO2

A number of means exist to capture carbon dioxide from gas streams, but they have not yet been optimised for the scale required in coal-burning power plants. The focus in the past has often been on obtaining pure CO_2 for industrial purposes rather than reducing CO_2 levels in power plant emissions.

Where there is carbon dioxide mixed with methane from natural gas wells, its separation is well proven. Several processes are used, including hot potassium carbonate which is energy-intensive and requires a large plant, a monoethanolamine process which yields high-purity carbon dioxide, amine scrubbing, and membrane processes.

A US hydrogen production plant uses vacuum swing adsorption gas separation technology for carbon capture.

In its Global Status of CCS 2021 report, the Global CCS Institute said that there were 27 commercial CCS projects in operation, though only one of these was in coal power generation (plus the Petra Nova plant that was placed in reserve shutdown in May 2020 – see below). The rest of these commercial facilities capture industrial emissions from natural gas processing units, chemical production, ethanol and steel plants, as well as fertiliser and hydrogen production. Altogther they capture about 40 Mt/yr. Global CCS publishes a facilities database.

Post-combustion capture

Capture of carbon dioxide from flue gas streams following combustion in air is much more difficult and expensive than from natural gas streams, as the carbon dioxide concentration is only about 14% at best, with nitrogen most of the rest, and the flue gas is hot. The main process treats carbon dioxide like any other pollutant, and as flue gases are passed through an amine solution

the CO_2 is absorbed. It can later be released by heating the solution. This amine scrubbing process is also used for taking CO_2 out of natural gas. There is a significant energy cost involved. For new power plants this is quoted as 20-25% of plant output, due both to reduced plant efficiency and the energy requirements of the actual process.

No commercial-scale power plants are operating with this process (the US Petra Nova plant has suspended CCS – see fuller mention below). At the 1300 MWe Mountaineer power plant in West Virginia which emits 8.5 Mt CO_2 annually, a $100 million Alstom pilot project successfully treated less than 2% of the plant's off-gas for CO_2 recovery, using chilled amine technology. There were plans to capture and sequester 20% of the plant's CO_2, but this was abandoned in 2011 due to lack of government support.

Oxyfuel combustion

Where coal is burned in oxygen rather than air, it means that the flue gas is mostly CO_2 and hence it can more readily be captured by amine scrubbing – at about half the cost of capture from conventional plants. A number of oxyfuel systems are operational in the USA and elsewhere, and the FutureGen 2 project involves oxy-combustion. Such a plant has an air separation unit, a boiler island, and a compression and purification unit for final flue gas.

The Integrated Gasification Combined Cycle (IGCC) plant is a means of using coal and steam to produce hydrogen and carbon monoxide (CO) from the coal and these are then burned in a gas turbine with secondary steam turbine (ie combined cycle) to produce electricity. If the IGCC gasifier is fed with oxygen rather than air, the flue gas contains highly-concentrated CO_2 which can readily be captured post-combustion as above.

Pre-combustion capture

Further development of the IGCC process will add a shift reactor to oxidise the CO with water so that the gas stream is basically just hydrogen and carbon dioxide, with some nitrogen. The CO_2 with some H_2S & Hg impurities are separated before combustion (with about 85% CO_2 recovery) and the hydrogen alone becomes the fuel for electricity generation (or other uses) while the concentrated pressurised carbon dioxide is readily disposed of. (The H_2S is oxidised to water and sulfur, which is saleable.) No commercial-scale power plants are operating with this process yet but see demonstration project sections below.

Currently IGCC plants typically have a 45% thermal efficiency.

Capture of carbon dioxide from coal gasification is already achieved at low marginal cost in some plants. One (albeit where the high capital cost has been largely written off) is the Great Plains Synfuels Plant in North Dakota, where 6 million tonnes of lignite is gasified each year to produce clean synthetic natural gas.

Oxy-fuel technology has potential for retrofit to existing pulverised coal plants, which are the backbone of electricity generation in many countries.

Storage & sequestration of carbon dioxide

Early CCS developments involved a single source linked to particular storage site. Economies of scale have led to a focus on hubs which aggregate, dehydrate and transport CO_2 streams from several sources. There are now about 15 hubs being set up. One of the most advanced hubs in development is the Northern Lights Project in the North Sea off Norway. It aggregates CO_2 streams, beginning with plants providing 0.8 Mt/yr of CO_2 and increasing to about 5 Mt/yr. Developed by Equinor, Shell and Total, the project will compress and liquefy CO_2 at source plants before transport by dedicated CO_2 ship to a storage site. The project is targeting a 2024 commissioning date.

There are three main categories of geological storage for CO_2: oil and gas replacement – notably enhanced oil recovery (EOR); coal seam storage; and deep saline aquifers. The first can have direct economic benefit offsetting the cost. The vast majority of storage potential is in deep saline aquifers. To 2014, 55 million tonnes of CO_2 had been sequestered with monitoring. At the end of 2016, 17 large-scale operational projects had a total potential capture rate of 30 Mt CO_2 per year, but only one-quarter of the captured CO_2 was being stored with appropriate monitoring and verification, according to the IEA's Energy Technology Perspectives. Most of this CO_2 is from natural gas processing.

In its Global Status of CCS 2020 report, the Global CCS Institute said: “Geological storage resources for CO_2 appear more than sufficient to meet global requirements under any net-zero emissions scenario. However, policy settings do not support a private business case for investment. Government funding of strategic storage resource appraisal programmes is essential.”

Enhanced oil recovery

Captured carbon dioxide gas can be put to good use, even on a commercial basis, for enhanced oil recovery (EOR), and the great majority of operating CCS projects are oriented thus. This is well demonstrated in west Texas, and today over 5800 km of pipelines connect oilfields to a number of carbon dioxide sources in the USA. The CO_2 acts to reduce the viscosity of the oil, enhancing its flow to recovery wells. It is then separated and re-injected.

At the Great Plains Synfuels Plant, North Dakota, the synthetic natural gas produced is passed through methanol at 70°C to remove compounds of sulfur and naphtha and capture most of the CO_2. Some 8000 tonnes per day of CO_2 gas is captured and piped 320 km into Canada for enhanced oil recovery at two oilfields. This represents about half of the CO_2 produced at full capacity. The Weyburn oilfield sequesters about 85 cubic metres of carbon dioxide per barrel of oil produced, a total of 19 million tonnes over the project's 20-year life. The first phase of its operation has been judged a success.

Chevron's Rangely project in the Rocky Mountain area injects 3 million tonnes of CO_2 per year supplied by pipeline for EOR in sandstone formations 1800 m deep. In Canada the Alberta Carbon Trunk Line captures 1.6 million tonnes of industrial CO_2 per year from an oil refinery and fertiliser plant for EOR.

Overall in the USA, over 6200 km of pipelines transport up to 72 million tonnes of CO_2 per year that the oil industry uses in enhanced oil recovery, 55 Mt from natural sources, 17 Mt anthropogenic. This produces 281,000 barrels of domestic oil per day, or 6% of US crude oil production. The EOR industry has captured, transported, and injected large volumes of CO_2 for oil recovery over four decades with no major accidents, serious injuries or fatalities. Present EOR technology has the potential to recover at least an additional 26 billion barrels of US oil, and improved technology could double this, while sequestering over 20 billion tonnes of CO_2. The USA in 2011 set up a National Enhanced Oil Recovery Initiative (NEORI) to help realize CO2-EOR's full potential as a national energy security, economic and environmental strategy. Its central recommendation is for a production tax credit for CO2 capture and sequestration with EOR.

In Texas, the Port Arthur demonstration project aims to capture 1 Mt/yr of CO2 from two steam methane reformers at Valero Energy Corporation's refinery, selling it for use in enhanced oil recovery. Another scheme separating CO_2 and using it for enhanced oil recovery is at In Salah, Algeria.

Depleted oil and gas fields

Total in France launched the first complete industrial-scale CCS chain in Europe in January 2010. CO_2 was captured from a 30 MWt boiler modified for oxyfuel combustion, piped 27 km, and injected into a depleted gas field 4500 metres deep. The Lacq basin pilot project in southwest France was intended to capture and store 120,000 tonnes of CO_2 over two years, but after three years 51,000 t was injected into the Rousse reservoir.

Coal seams

Injecting carbon dioxide into deep, unmineable coal seams where it is adsorbed to displace methane is another potential use or disposal strategy. The displacement effect means that coal seam CO_2 injection could be most effective as part of the commercial production of coal seam methane (also known as coal bed methane, effectively: natural gas), an increasingly important and relatively new energy source.

Storage in coal seams is different since the CO_2 is adsorbed in the coal matrix instead of being held within the pores of the rocks as in saline aquifers and oil-gas systems. The properties of the coal strongly influence whether CO_2 will adsorb into it. Currently the economics of enhanced coal bed methane extraction with CO2 disposal are often not as favourable as with enhanced oil recovery, but the potential is large as coal seam gas is increasingly tapped.

Saline aquifers

These hold most potential. They are underground formations of deep porous sedimentary rock such as sandstone, that are saturated with salty water which is unfit for human consumption or agricultural use, and covered by a layer of impermeable cap rock (such as shale or clay), which acts as a seal. Once injected into the formation, the CO_2 dissolves into the saline water in the reservoir rock. CO_2 storage in deep saline formations usually takes place at depths below 800 metres. At this depth, the CO_2 will be at high enough pressures to remain in a liquid-like state.

The world's first industrial-scale CO_2 storage was at Norway's Sleipner gas field in the North Sea, where about one million tonnes per year of compressed liquid CO_2 separated from methane is

injected into a deep reservoir (saline aquifer) about a kilometre below the sea bed and remains safely in place. The $80 million incremental cost of the sequestration project was paid back in 18 months on the basis of carbon tax savings at $50/tonne. (The natural gas contains 9% CO_2 which must be reduced before sale or export.) The Utsira sandstone formation there, about one kilometre below the sea bed, is said to be capable of storing 600 billion tonnes of CO_2. To 2017, over 17 million tonnes of CO_2 had been stored over 20 years at about 1.0 Mt/yr. In 2007 the Snohvit project, with 8% reservoir CO_2 content, joined Sleipner in CCS there, with 0.7 Mt/yr capacity.

West Australia's **Gorgon** natural gas project producing 15.6 million tonnes of LNG per year taps natural gas with 14% CO_2. Capture and geosequestration of this is expected to reduce the project's total emissions from 6.7 to 4.0 million tonnes of CO_2 per year. From 2020 the project will have capacity for 3.4 to 4.0 million tonnes per year of pressurised supercritical CO_2 to be injected into the Dupuy formation – a saline aquifer 2500 metres deep below Barrow Island. In each of 2017-18 and 2018-19 the project released 9 Mt CO_2 and by August 2021 it had sequestered 5 Mt CO_2 in total. This is the world's largest commercial-scale CO_2 injection facility and the Gorgon joint venture is investing approximately A$2.5 billion in the design and construction of it. The water content of the CO_2 required modification of the compressors, and caused delays in commissioning. The Australian government committed $60 million to the Gorgon Carbon Dioxide Injection Project as part of the Low Emissions Technology Demonstration Fund. Gorgon is 47.3% owned by Chevron.

Some CO_2 injection to saline aquifers involves acid gas, disposing of hydrogen sulphide and CO_2 separated for a natural gas stream. Chevron's Acheson Field in Canada was one of the first to use this acid gas injection.

Saline formations have the largest storage potential globally and a number of CO_2 storage demonstration projects are proving their effectiveness to maximise storage capacity and containment.

R&D general

The scale of envisaged need for CO_2 disposal far exceeds today's plans. Safety and permanence of disposition are key considerations in sequestration.

In Australia the CO2CRC Otway project injected over 80,000 t CO_2 into a depleted gas reservoir in Victoria. This is subject to an intensive monitoring programme.

Research on geosequestration is ongoing in several parts of the world. The main potential appears to be deep saline aquifers and depleted oil and gas fields. In both, the CO_2 is expected to remain as a supercritical gas for thousands of years, with some dissolving.

In 2016 it was reported that Iceland had trialled pumping CO_2 and water into hot underground rocks and turning it into limestone over about two years.

Large-scale storage of CO_2 from power generation will require an extensive pipeline network in densely populated areas. This has safety implications.

Given that rock strata have held CO_2 and methane for millions of years there seems no reason that carefully-chosen chosen ones cannot hold sequestered CO_2. However, the eruption of a million tonnes of CO_2 from Lake Nyos in Cameroon in 1986 asphyxiated 1700 people, so the consequences of major release of heavier-than-air gas are potentially serious.

Producing oxygen for oxyfuel and IGCC

Today most oxygen is recovered cryogenically from liquid air, which is a relatively expensive process.

The main prospective means of economically producing large amounts of oxygen is the ion transport membrane (ITM) process. It is being developed for use in feeding integrated gasification combined cycle (IGCC), oxyfuel combustion, and other advanced power generation systems including underground coal gasification. In the USA, EPRI is involved on behalf of the electric utilities in helping to scale-up ITM technology for clean energy.

ITM technology uses a ceramic material which, under pressure and temperature, ionizes and separates oxygen molecules from air. No external source of electrical power is required. Relative to traditional cryogenic air separation units, ITM technology could decrease internal power demand by as much as 30% and capital costs by approximately 30% in the oxygen supply systems at oxyfuel and IGCC power plants.

The oxygen requirements for the power generation industry could grow substantially in supporting advanced coal-based power

generation and integrated carbon capture technology. EPRI estimates the current US power generation industry share of the oxygen market is about 4%, but it could become the dominating market driver, accounting for more than 60% of the future market, or approximately two million tones per day of oxygen by 2040.

Carbon capture and utilization/use (CCU)

Obviously enhanced oil recovery outlined above amounts to utilization as well as storage, hence CCS, but beyond that the CO_2 may be used with hydrogen to make methanol, which is a plausible substitute for petrol/gasoline, and also dimethyl ether from that, a good diesel substitute. There are other possibilities for embedding the carbon in materials such as polycarbonates, which are long-lasting. However, the CO_2 quantities involved are trivial compared with the accepted need to reduce carbon emissions. For further information, see the information page on Hydrogen Production and Uses.

Economics

The World Coal Institute noted that in 2003 the high cost of carbon capture and storage (estimates of $150-220 per tonne of carbon, $40-60/t CO_2 – 3.5 to 5.5 c/kWh relative to coal burned at 35% thermal efficiency) made the option uneconomic. But a lot of work is being done to improve the economic viability of it, and the US Department of Energy (DOE) was funding R&D with a view to reducing the cost of carbon sequestered to $10/tC (equivalent to 0.25 c/kWh) or less by 2008, and by 2012 to reduce the cost of carbon capture and sequestration to a 10% increment on electricity generation costs. These targets now seem very unrealistic.

A 2000 US study put the cost of CO_2 capture for IGCC plants at 1.7 c/kWh, with an energy penalty 14.6% and a cost of avoided CO_2 of $26/t ($96/t C). By 2010 this was expected to improve to 1.0 c/kWh, 9% energy penalty and avoided CO_2 cost of $18/t ($66/t C), but these numbers now seem unduly optimistic.

Figures from IPCC Mitigation working group in 2005 for IGCC put capture and sequestration cost at 1.0-3.2 c/kWh, thus increasing electricity cost for IGCC by 21-78% to 5.5 to 9.1 c/kWh. The energy penalty in that was 14-25% and the mitigation cost $14-53/t CO_2 ($51-200/tC) avoided. These figures included up to $5 per tonne

CO_2 for transport and up to $8.30 /t CO_2 for geological sequestration.

In 2009 the OECD's International Energy Agency (IEA) estimated for CCS $40-90/t CO_2 but foresees $35-60/t by 2030, and McKinsey & Company estimated €60-90/t reducing to €30-45/t after 2030.

ExxonMobil is proposing that, where amine scrubbing is employed, the whole power plant exhaust is directed to a carbonate fuel cell which will generate over 20% more power overall, instead of costing 10% of the power due to diversion of steam. The CO_2 still needs to be disposed of.

A 2017 study by Energy Innovation in the USA comparing ultrasupercritical coal with and without CCS (90% capture) showed that the LCOE figures were $151.34/MWh and $92.46/MWh, respectively – nearly two-thirds more. This was attributed largely to the extra energy required to extract, pump, and compress the CO_2, and hence not amenable to great improvement.

FutureGen demonstration projects, USA

About 2005 the DOE announced the $1.3 billion FutureGen project to design, build and operate a nearly emission-free coal-based electricity and hydrogen production plant. Some $250 million of the funding was to be provided by industry, from about eight companies. The FutureGen initiative would have comprised a coal gasification (IGCC) plant with additional water-shift reactor, to produce hydrogen and carbon dioxide. It would also involve development of the ITM oxygen separation technology. About 700,000 tonnes of CO_2 (some 60% of throughput) per year would then be separated by membrane technology and sequestered geologically. The hydrogen would have been be burned in a 275 MWe generating plant and in fuel cells. Later FutureGen figures referred to 90% CO_2 capture and 330 MWe gross, 240 MWe net generation.

Construction of this original FutureGen was due to start in 2009, for operation in 2012, with target of 90% carbon capture. The project was designed to validate the technical feasibility and economic viability of near-zero emission coal-based generation. In particular it aimed to produce electricity with only a 10% cost premium and to show that hydrogen can be produced at $3.80 per GJ, equivalent to petrol at 12.7 cents per litre. In December 2007 Mattoon Illinois was chosen as the site of the demonstration plant. However, in January 2008 the DOE announced that it would withdraw its funding for the

project, expressing concerns over escalating costs – its 74% share having doubled to $1.3 billion. The Mattoon site in Coles County was subsequently sold.

Under the new Administration in 2010 however, the project was reconsidered, and design work, geological investigations and a revised cost estimate proceeded. In August 2010 DOE said that it would abandon the original FutureGen idea and would now retrofit unit 4 of Ameren's existing oil-fired plant in Meredosia, Illinois, with oxy-combustion rather than IGCC, calling this FutureGen 2.0 – "a clean-coal repowering program and carbon dioxide storage network." It would burn pulverised coal and capture over 90% of the CO_2 produced (1.3 Mt/yr over 30 years), to produce 166 MWe net. A pipeline would link it to a regional CO_2 storage hub, and a site will be sought for this to enable sequestration in the Mt Simon Formation. Ameren would use B&W technology for oxy-combustion repowering of the plant, and FutureGen Alliance will focus on the pipeline and storage, with a view to also drawing on other CO_2 sources within 160km, so that some 500 million tonnes capacity was sought.

The DOE said that it would be prepared to contribute $1.1 billion of the $1.65 billion cost to it as a public-private partnership involving the FutureGen Industrial Alliance (FGA), Ameren Energy Resources, Babcock & Wilcox, and Air Liquide Process & Construction, Inc. Late in 2010 members of the FGA included domestic coal companies Peabody, BHP Billiton, Rio Tinto and Consol Energy, plus E.On. No domestic utilities remained, though Exelon had indicated an intention to join. In December 2012, the Illinois Commerce Commission mandated that Commonwealth Edison (ComEd) and Ameren Illinois had to purchase the electricity from the project for 20 years, but the utilities challenged this on the grounds of cost.

After identifying a suitable sequestration site in Morgan County, the design phase of the project was announced in February 2013. Construction was due be completed in 2015, with the project being on line mid-2016, but this was delayed as most members of the FGA dropped out, leaving only Peabody, Glencore and Anglo American. In February 2015 DOE cancelled further funding for the project, after having spent $202 million on it.

Other demonstration projects

North America

The US Department of Energy (DOE) has said that funding would be made available to assist other projects that aim to add carbon capture and storage (CCS) to existing coal plants, but will no longer include hydrogen production as part of the project. Over half of the CO_2 capture projects in development or operation globally are in North America, and all but one of these is oriented to provide CO_2 for enhanced oil recovery (EOR).

Duke Energy Corp in the USA is building a $3 billion, 618 MWe, IGCC plant at Edwardsport, Indiana ($4850/kW). This is a regulated plant, but Duke says that consumers will not be asked to pay for more than $2.72 billion of its final construction cost, excluding financing.

In Texas, the Petra Nova project near Houston, a partnership of the US DOE, NRG Energy and JX Nippon, is set up to capture 1.4 million tonnes of CO_2 per year (90%) from NRG's WA Parish 240 MWe power plant and use it for enhanced oil recovery. In a post-combustion process the flue gas is cooled and the CO_2 removed by amine scrubbing. The CO_2 is released from the solvent with low-pressure steam. The Petra Nova Parish plant started up late in 2016 on time and on budget, and is the largest post-combustion carbon capture project installed on an existing coal-fuelled power plant.

By April 2017 Petra Nova had delivered 300,000 tonnes of CO_2 through a 30 km pipeline to the West Ranch oilfield to increase oil production from 300 to 15,000 barrels per day. The system captures more than 90% of carbon emissions from a 240 MW equivalent stream of flue gas. It is rated at 4776 tonnes of CO_2 captured daily, effectively 1.4 million tonnes per year. The plant is reported to cover costs through the economic benefit of enhanced oil recovery. Due to the low oil price, carbon capture paused in March 2020 but NRG planned to restart it when economic conditions permit. The $1 billion plant was financed with loans from the Japan Bank for International Cooperation and Mizuho Bank, supported by Nippon Export and Investment Insurance. The project also obtained $167 million in grants from the US DOE's Clean Coal Power Initiative programme.

Summit Power Group's Texas Clean Energy Project (TCEP) at Penwell was to be a 400 MWe IGCC power plant burning coal with CCS, capturing 90% of CO_2 and 90% of NO_x pre-combustion. It had $811 million funding from the DOE Clean Coal Power Initiative

towards its $4 billion cost. It was due to operate from 2019, but in 2016 DOE withdrew funding. Of the approx 2 million t/yr of CO_2 captured, 83% would be used for enhanced oil recovery in the West Texas Basin. Of the 400 MWe, 106 MW would be used to run the major project equipment on site, 16 MW would be used to compress CO_2, and 42 MW would be used to produce urea, leaving 236 MWe for the grid. In December 2015 Summit signed an engineering, procurement and construction (EPC) contract with China Huanqui Contracting & Engineering Corporation (HQC), a subsidiary of China National Petroleum Corporation (CNPC), and SNC-Lavalin. However, the project appears to be shelved.

Mississippi Power and Southern Company's Kemper County Energy Facility in Mississippi was due to start up in 2016, but costs have blown out from $2.9 billion to over $7.5 billion. It was to be a 580 MWe lignite-fired plant with gasification for IGCC and CCS. In June 2017 the company suspended start-up and operations activities involving the lignite gasification portion of the project in response to an order from the Mississippi Public Service Commission to remove financial risk to ratepayers over “unproven technology”. Though the company said that the lignite gasification part of the plant worked, the combined-cycle plant was fully converted to operate using natural gas, as it has since 2014.

The Kemper plant aimed to gasify lignite using two transport integrated gasification (TIG) units and burn the syngas, principally hydrogen, using IGCC to generate 582 MWe of electricity, then capture 65% of the CO_2 – 3 Mt/yr – which would be sold for enhanced oil recovery. Southern Company planned to pass on $4.2 billion in costs to (its subsidiary) Mississippi Power ratepayers, but in mid-2017 announced that it would absorb $5.87 billion in losses on the project. The lignite gasification plant is not expected to be competitive with gas prices below $5 per million BTU. Project ECO_2S aims to establish a one billion tonne CO_2 storage site on a 12,000-hectare site near the Kemper County Energy Facility which could serve as a regional CO_2 storage hub for both power and industrial CO_2 point sources.

Net Power, backed by Toshiba, Exelon and others, has built and commissioned a 50 MWt plant in La Porte, Texas with oxy-fuel combustion of natural gas, and hot recycled CO_2 driving a turbine ('Allam-Fetvedt cycle') rather than going up the stack. Apart from electricity, the plant produces only water and pipeline-ready CO_2. In contrast to normal CCGT, no steam is involved and CO_2 removal is integral. In May 2018, the project achieved first firing of a commercial-scale combustor, which was made by Toshiba Energy Systems & Solutions, validating the “fundamental operability and

technical foundation" of this project. Thermal efficiency of 59% is claimed, compared with 48% for CCGT with carbon capture. The CO_2 is used for enhanced oil recovery. Net Power plans to bring a 300 MWe Allam cycle plant online in Texas in 2022.

In its *Global Status of CCS 2020 report*, the Global CCS Institute said: "This technology can produce electricity with more than 97 per cent CO_2 capture at a levelised power price approximately 22 per cent higher than today's conventional natural gas combined cycle. The cost premium is expected be under 10 per cent by 2050."

In New Mexico, Enchant Energy is developing a CCS project for its coal-fired San Juan power station. Up to 6 million t/yr of CO_2 from post-combustion capture will be used for EOR in the Permian Basin from about 2023.

In Canada, the 110 MWe unit 3 of the Boundary Dam lignite-fired plant is the world's only commercial-scale CCS power station, operating from early 2015. SaskPower's Boundary Dam unit 3 plant was rebuilt in 2013 and came online in October 2014. Prior to upgrading it at a cost of C$1.47 billion, with C$900 million of this for the CCS system, it operated at 139 MWe and released 3604 tCO_2 per day. Reduced to 110 MWe it now releases 354 t/day, so captures about 90%, about 700,000 tonnes per year from flue gas. By late 2020 it had captured and stored 3.6 Mt CO_2. An April 2016 Parliamentary Budget Office report found that CCS at Boundary Dam doubles the price of electricity. Most of the CO_2 is sold at about $25/t for EOR at the Weyburn oilfield, some is piped to nearby Aquistore Project for geological storage. SaskPower is not installing CCS on units 4&5, which are due to be shut down in the 2020s.

The Quest project in Canada's oil sands commissioned in 2015 captures up to 1 million tonnes of CO_2 per year from hydrogen production at the Scotford Oil Sands Upgrader for storage at a depth of about 2 km in an onshore saline aquifer.

In Canada, the Alberta Carbon Trunk Line started operation in March 2020 as the world's largest CO_2 transport infrastructure. Its capacity is 14.6 million t/yr. So far it takes 1.6 Mt/yr CO_2 from the Sturgeon oil refinery and the Nutrien fertiliser plant for EOR in central Alberta.

Europe

In 2007, EU leaders endorsed a European Commission plan for up to 12 CCS demonstration power plants by 2015. By the end of

2018 there were no such plants, nor any plans, though €1 billion is reported to have been spent. CCS was also promoted by the International Energy Agency (IEA) and the Intergovernmental Panel on Climate Change (IPCC) as a promising means of transition to a low-carbon economy.

The CCPilot 100+ project at Ferrybridge in West Yorkshire, UK, has commenced operation. It can capture 100 tonnes of CO_2 per day from 5 MWe of coal-fired power plant. The operator is Scottish & Southern Energy in collaboration with Vattenfall and Doosan Babcock. It uses a post-combustion amine scrubbing process and is subsidised by the government. Vattenfall expected to apply experience from it to a much larger demonstration plant at Jaenschwalde in Germany, to operate from 2015, but cancelled the project at the end of 2011.

In the UK a competition was launched by the UK government in 2007 to support a coal-fired power plant demonstrating the full chain of CCS technologies (capture, transport, and storage) on a commercial scale. The winning project bid would have to demonstrate post-combustion capture (including oxyfuel) on a coal-fired power station, with the carbon dioxide being transported and stored offshore. The project would have to capture around 90% of the CO2 emitted by the equivalent of 300MW-400MW generating capacity. The successful project bid should demonstrate the entire CCS chain by 2014. A further CCS commercialisation competition was launched in 2012, and in 2013 the government selected two proposals, the White Rose Project and the Peterhead project, which were both funded to undertake engineering work. It was envisaged that the companies would make final investment decisions about the end of 2015, with the government taking funding decisions then. However, in November 2015 the government announced withdrawal of funding for both projects.

Scottish & Southern Energy and Shell UK agreed in November 2011 to build the first commercial-scale CCS demonstration project at Peterhead in the northeast of Scotland, at a cost of £1 billion. The project aimed to design and develop a full chain, post-combustion CCS facility which would capture 85% to 90% of the CO2 from one 385 MW combined cycle gas turbine unit at SSE's Peterhead Power Station. It was planned that the 1 million tonnes per year of CO2 would then be pumped 100 km to Shell's Goldeneye gas field 2.5 km beneath the North Sea which will have ceased production using, as far as possible, existing infrastructure. Some £1bn from the EU and UK government was expected for the project. Earlier Scottish Power cancelled plans for a similar project at Longannet coal-fired power station. Shell said it remained committed to CCS.

Capture Power Ltd had applied for planning permission for a 448 MWe coal-fired unit with oxyfuel combustion and CCS at the Drax power station in Yorkshire. This White Rose project was designed to capture 90% of carbon emission, about 2 million tonnes per year, which would be piped by National Grid to to a 'saline rock formation' below the North Sea. The EU in 2014 provided €300 million funding.

In Denmark a pilot project at the 420 MWe Elsam power plant is capturing CO_2 from post-combustion flue gases under the auspices of CASTOR (CO_2 from Capture to Storage). Flue gases are passed through an absorber, where a solvent captures about 90% of the CO_2. The pregnant solution is then heated to 120°C to release pure CO_2 at the rate of about one tonne per hour for geological sequestration. Cost is expected to be €20-30 per tonne.

Zero Emission Platform (ZEP), founded in 2005, is a European CCS lobby group involving major utilities. ZEP serves as advisor to the European Commission on the research, demonstration and deployment of CCS. However, early in 2015 RWE, Vattenfall and Gas Natural Fenosa (also apparently EdF) dropped out, pleading restricted time and budget and saying that "we do not have the necessary economic framework conditions in Europe to make CCS an attractive technology to invest in." In mid-2021 ZEP listed a large number of "market-ready" projects awaiting investment.

In the Netherlands, the Rotterdam Opslag and Afvang Demonstratieproject (ROAD, Rotterdam Capture and Storage Demonstration Project) set up by E.On and GdF Suez/ Engie aimed to capture 1.1 Mt CO_2/yr from a 250 MWe coal-fired plant at the Maasvlakte-3 plant in Rotterdam. However, it was abandoned in mid-2017, leaving no other European CCS power projects.

In Germany, Vattenfall's Schwarze Pumpe lignite power plant started a 30 MW pilot CCS project in 2007, but never proceeded past this and the company will abandoned the project in 2018. In 2014 Vattenfall announced it would abandon CCS altogether.

Rest of world

In China, the first phase of Huaneng Group's $1.5 billion GreenGen project – a 250 MWe oxyfuel IGCC power plant burning syngas (mainly hydrogen and carbon monoxide) from coal feed – commenced operation at Tianjin in 2012 and has been fully operational since 2014. The second phase involves a pilot plant which draws about 7% of the syngas from the IGCC power plant,

shifts CO and water to CO_2 and H_2, then separates the CO_2 from the H_2 after desulphurisation, and produces electricity from hydrogen. The 60,000 to 100,000 tpa CO_2 is used for enhanced oil recovery. Phase 3 will be a 400 MWe commercial IGCC plant with CCS to capture up to 2 million tonnes of CO_2 per year, from sometime in the 2020s.

In China, the major utility China Datang teamed up with Alstom to build two demonstration CCS projects. A 2x350 MWe coal-fired plant at Daqing, Heilongjiang province, would be equipped with Alstom's oxy-firing technology, and a 2x1000 MWe ultra-supercritical coal-fired plant at Dongying, Shandong province, would use Alstom's post-combustion capture technology, either chilled ammonia or advanced amines. The two projects were expected to each capture over one million tonnes of CO_2 per year, which would be about 40% of output from Daqing and 15% from Dongying, though Alstom says that the actual levels of capture and storage have not yet been defined and will be in the scope of the first feasibility studies of the respective projects. Adjacent oilfields would be used for sequestration, enabling enhanced oil recovery (EOR). Both projects were shelved in 2016-17, though construction had begun on the Dongying plant in 2016.

The Sinopec Shengli power plant CCS project is planned to come online in the 2020s, with post-combustion capture and 1 Mt/yr CO_2 used for EOR.

The Shanxi International Energy Group is planning a CCUS project for a new supercritical coal-fired power plant with oxy-fuel combustion at Taiyuan in Shanxi province to come online in the 2020s and capture 2 Mt/yr CO_2.

The Uthmaniyah project in the Eastern Province of Saudi Arabia commissioned in 2015 captures around 800,000 tonnes of CO_2 per year from the Hawiyah natural gas liquids recovery plant to be injected for enhanced oil recovery (EOR) at the Ghawar oilfield.

In Australia the $240 million Callide Oxyfuel project in Queensland aims to demonstrate oxyfuel capture technology retrofitted to a 30 MW unit of an existing coal-fired power plant and to research how it might be applied to new power stations. The plant was commissioned in 2012 and was to run for an extended test period until November 2014. By mid-2013 the project had demonstrated CO_2 capture rates from the oxyfuel flue gas stream to the CO_2 capture plant in excess of 85%, and produced a high quality CO_2 product suitable for geological storage. The project achieved more than 10,000 hours of oxy-combustion and more than 5,000 hours

of carbon capture from Callide A. The plant was then decommissioned. CS Energy led the project and is working closely with an international team of partners including IHI Corporation (Japan), J-Power (Japan), Mitsui & Company (Japan), and Xstrata Coal.

Also in Australia the $150 million Delta Post Combustion Capture project hosted at Delta's 1320 MWe Vales Point coal-fired power station in NSW aimed to demonstrate capture and sequestration of 100,000 t/yr of CO_2 by 2015. However, after massive losses the plant was sold for a token sum in November 2015, with no mention of the CCS project.

Both Australian projects were funded by federal and state governments and the coal industry.

Gasification processes

In conventional plants coal, often pulverised, is burned with excess air (to give complete combustion), resulting in very dilute carbon dioxide at the rate of 800 to 1200 g/kWh.

Gasification converts the coal to burnable gas with the maximum amount of potential energy from the coal being in the gas.

In Integrated Gasification Combined Cycle (IGCC) the first gasification step is pyrolysis, from 400°C up, where the coal in the absence of oxygen rapidly gives carbon-rich char and hydrogen-rich volatiles.

In the second step the char is gasified from 700°C up to yield gas, mostly CO, leaving ash. With oxygen feed, the gas is not diluted with nitrogen.

The key reactions today are $C + O_2$ to CO, and the water gas reaction: $C + H_2O$ (steam) to CO & H_2 – syngas, which reaction is endothermic.

In gasification, including that using oxygen, the O_2 supply is much less than required for full combustion, so as to yield CO and H_2. The hydrogen has a heat value of 121 MJ/kg – about five times that of the coal, so it is a very energy-dense fuel. However, the air separation plant to produce oxygen consumes up to 20% of the gross power of the whole IGCC plant system. This syngas can then be burned in a gas turbine, the exhaust gas from which can then be used to raise steam for a steam turbine, hence the "combined cycle" in IGCC.

To achieve a much fuller clean coal technology in the future, the water-shift reaction will become a key part of the process so that:

- $C + O_2$ gives CO, and
- $C + H_2O$ gives CO & H_2, then the
- $CO + H_2O$ gives CO_2 & H_2 (the water-shift reaction).

The products are then concentrated CO_2 which can be captured, and hydrogen. (There is also some hydrogen from the coal pyrolysis), which is the final fuel for the gas turbine.

Overall thermal efficiency for oxygen-blown coal gasification, including carbon dioxide capture and sequestration, is about 73%. Using the hydrogen in a gas turbine for electricity generation is efficient, so the overall system has long-term potential to achieve an efficiency of up to 60%.

Present trends

The clean coal technology field is moving in the direction of coal gasification with a second stage so as to produce a concentrated and pressurised carbon dioxide stream followed by its separation and geological storage. This technology has the potential to provide what may be called "zero emissions" – in reality, extremely low emissions of the conventional coal pollutants, and as low-as-engineered carbon dioxide emissions.

This has come about as a result of the realisation that efficiency improvements, together with the use of natural gas and renewables such as wind will not provide the deep cuts in greenhouse gas emissions necessary to meet future national targets.

The US DOE sees "zero emissions" coal technology as a core element of its future energy supply in a carbon-constrained world. It had an ambitious program to develop and demonstrate the technology and have commercial designs for plants with an electricity cost of only 10% greater than conventional coal plants available by 2012, but this is at least postponed.

Australia is very well endowed with carbon dioxide storage sites near major carbon dioxide sources, but as elsewhere, demonstration plants will be needed to gain public acceptance and show that the storage is permanent.

Natural gas as alternative fuel

There are many advocates for the use of natural gas as an alternative to coal for electricity generation, on the grounds that it emits much less CO_2 per kWh generated. This is true on almost any basis of comparison, but it ignores the global warming potential of leaked natural gas, and the CO_2 emissions in transporting it as LNG (up to one-third of the energy is consumed in transport). Leakage of 3% of the natural gas will bring it into approximate parity with coal-fired electricity in terms of global warming effect.

There is a range of ways of using natural gas primarily for power generation:

Central Heat and Power (CHP) – Typically burn in a combined cycle gas turbine (CCGT) for electricity, using exhaust gas to heat steam boiler to make more electricity, and finally using the exhaust stream to heat buildings or other purposes. Thermodynamic efficiencies of 80% for this have been reported.

Combined cycle gas turbine (CCGT) – primary gas turbine with secondary steam turbine, with 55-60% thermal efficiency.

Open cycle gas turbine – 35-40% efficiency, or straight steam boiler with about 40% efficiency; favoured for backing up intermittent renewables, due to quick start.

All of these have potential for CCS.

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